ElderCare – Emergency Assist

Libraries, APIs & Dependencies

This document lists the libraries, browser APIs, external resources, and runtime requirements used by the current ElderCare browser prototype. It distinguishes actual dependencies from services that are not yet integrated.

1. Project Technology Overview

Area	Technology used	Dependency type
Page structure	HTML5	Browser standard
Styling and layout	CSS3	Browser standard
Application logic	Vanilla JavaScript	Browser runtime
Voice input	Web Speech API – SpeechRecognition	Browser API; support varies
Location	Geolocation API	Browser API; permission required
Password hashing	Web Crypto API – SHA-256	Browser API; secure context requirements may apply
Local persistence	Web Storage – LocalStorage	Browser API
Cross-tab updates	Web Storage storage event	Browser API
Spoken prompts	Speech Synthesis API	Browser API
Audio feedback	Web Audio API	Browser API
Map display/opening	Google Maps URL links	External web destination
Typography	Google Fonts – Playfair Display and Lora	External font resource

2. Browser APIs and Their Purpose

Web Speech API – SpeechRecognition

Captures spoken input and converts it to text for configured emergency-phrase checks. Requires microphone permission and a browser that supports SpeechRecognition; behavior can vary by browser.

Geolocation API

Requests the device's location and can monitor location updates. The user must grant location permission; browser security and device settings can affect availability.

Web Crypto API

Provides cryptographic functions used by the prototype for SHA-256 password hashing. SHA-256 is a hash function, not a password-specific key-derivation algorithm.

LocalStorage API

Stores prototype application data in the browser under the key ecv2. Data is local to that browser profile and is not automatically synchronized to other devices.

Storage event

Notifies other same-origin browser tabs/windows when local storage changes, allowing the interface to refresh across tabs.

Speech Synthesis API

Produces spoken prompts or feedback using voices available from the browser/device.

Web Audio API

Generates audio cues and alert sounds in the browser.

3. External Resources

Google Fonts: Playfair Display and Lora are loaded as external font resources. An internet connection may be needed for the fonts to load; fallback fonts should be used if unavailable.

Google Maps: The application creates or opens Google Maps URLs for location viewing. This is URL-based integration, not a bundled JavaScript Maps SDK or a configured Maps API key.

4. Third-Party JavaScript Libraries and Package Dependencies

The inspected prototype is implemented with plain HTML, CSS, and Vanilla JavaScript. No third-party JavaScript framework or package dependency is required by the current implementation. There is no indicated use of React, Angular, Vue, Bootstrap, jQuery, or a Node.js package installation.

Package manager: No npm/yarn/pnpm dependency installation is required for the current standalone browser prototype. **Build step:** No bundler or compilation step is required to run the current HTML file.

5. Runtime and Browser Requirements

- Modern web browser with JavaScript enabled.
- Microphone access for voice recognition; SpeechRecognition availability differs between browsers.
- Location permission for GPS/location features.
- Secure context (normally HTTPS, or localhost during development) for browser features that require it.
- Internet access for externally hosted Google Fonts and for opening Google Maps links.
- User interaction/permissions may be required for audio playback, speech, microphone, and location features.

6. Database and Backend Dependencies

A separate SQLite database schema has been prepared, but it is not connected to the current website. The current prototype uses browser LocalStorage. Therefore, SQLite is a prepared database artifact, not an active runtime dependency of the present HTML application.

A Python Flask backend, hosted database, API server, cloud synchronization, SMS/email provider, and pharmacy service are not currently integrated. They would be additional dependencies if added in a future version.

7. Setup Notes

- Keep the main HTML file and any local assets in the project repository.
- Open the project through a local development server or deploy it to an HTTPS-enabled static host for better compatibility with browser permissions.
- Grant microphone and location permissions when testing voice and GPS features.
- Test voice recognition in the target browser before demonstrating the application.
- Do not describe browser-opened SMS/email/call links as automatic message delivery or background emergency calling.

Dependency status: The current application is a standalone browser-side prototype. Its main dependencies are browser-provided Web APIs and two externally loaded Google Fonts; no third-party JavaScript package manager dependencies are required.